Cars 2 Has an Alibi

Class 17 Instructor Notes | 50 minutes of a 75-minute period | Four-page handout | Laptops, teams of 2–3

Placement and purpose

First class covering bivariate data. Runs on the day the Pennypack mini-project is due at midnight, so the activity deliberately uses **no** Pennypack data — students are still writing that up, and nothing here should preempt their interpretation.

The one idea: r measures one specific thing — how tightly the points hug a straight line — and "the most unusual point" is a different question with a different answer. Part 1 builds the univariate tools, Part 2 derives r from standard units, Part 3 shows that the obvious outlier and the point that most moves r are not the same film, and why.

The spine. The box plot in Part 1 flags exactly one flier — Cars 2 — and the scatter plot in Part 2 makes it look even worse. In Part 3 students find that removing it **raises** r, and that the largest single effect on r belongs to Onward instead. Cars 2 is not innocent; it has an alibi for this particular charge. Do not spoil this.

Sets up Class 18. Part 2 has students derive r as the mean of the product of standard units by hand. That is the same derivation as slides 9–15 of the Class 18 deck, so those can compress to a single recap slide — and slide 30 ("in standard units the slope is r and the intercept is zero") lands as a payoff rather than a new fact.

Timing

18 min	Part 1 — `stats()`, quartiles by hand, box plot
18 min	Part 2 — scatter, standard units, r
14 min	Part 3 — the alibi, then the four data sets

Leaves 25 minutes for practice quiz questions, Pennypack tips, and correlation vs. causation.

If you are behind: run Q3.5 as a demo from the front rather than cutting it — the four plots take thirty seconds to show and the discussion can happen out loud. Q3.3 is the one that needs quiet working time.

Code blanks

Only two, both in Part 2:

```
return (any_numbers - np.mean(any_numbers)) / np.std(any_numbers)
return np.mean(standard_units(x) * standard_units(y))
```

18 min — Part 1. One Variable at a Time

Twenty-one films survive the `dropna()`.

	mean	median	mean − med	IQR
`rotten_tomatoes`	88.52	96	**−7.48**	17.0
metacritic	79.38	81	−1.62	21.0
`critics_choice`	87.14	89	−1.86	12.0

Q1.2: Rotten Tomatoes. A mean well below the median means a long left tail — a few very low values dragging the average down while most films cluster high.

The reason is the measurement scale, not the movies. Rotten Tomatoes reports the *percentage of critics who were positive*, so it saturates: 14 of the 21 films sit at 90 or above and it cannot exceed 100. Metacritic averages numerical scores and does not pile up at the ceiling. Worth saying out loud — students assume the difference says something about Pixar.

Q1.3 (by hand): Q1 = 26.5, Q2 = 36, Q3 = 51, IQR = 24.5. Same twelve numbers as the MathBits slide, so it doubles as a check that the lecture definition landed.

Q1.4: Rotten Tomatoes most favorable, metacritic least. **One** flier in the entire plot, on `rotten_tomatoes`.

Q1.5: Cars 2 — 40 / 57 / 67. Expect teams to say yes, obviously an outlier. Let them.

20 min — Part 2. Two Variables at a Time

Q2.2: Standardizing changes the axis numbers and centers the cloud on zero. It does not change the shape of the pattern at all — because subtracting a constant and dividing by a constant is just relabeling the ruler.

Q2.3 sign table: both above average, or both below → **positive** product. One above and one below → **negative**. So when the variables track each other, nearly every product is positive and they add up; when they do not, positives and negatives cancel toward zero.

Quadrant count (metacritic vs. `critics_choice`): **17 of 21** films sit in the two + quadrants (11 upper-right, 6 lower-left). The 4 in − quadrants — Monsters, Inc., Cars, Finding Dory, Incredibles 2 — all sit within about 0.5 SD of the origin, so their negative products are tiny. Worth pointing out: *where* a point sits matters as much as *which* quadrant.

Expect confusion if students treat the table as a data-entry task. The 2×2 is conceptual (signs only); the quadrant question is what ties it to the data. The notebook now draws the zero lines on the standardized plot and shows film names beside the `_su` columns, since `rmc` dropped them.

	`rotten_tomatoes`	**metacritic**
metacritic	**0.802**	
`critics_choice`	**0.852**	**0.865**

Q2.4: metacritic and `critics_choice` agree most closely ($r = 0.865$). All three are strong and positive, which is the unsurprising part — they are all rating the same films.

14 min — Part 3. The Alibi

Q3.1: r = **0.802** with all 21 films, **0.826** with Cars 2 removed. Removing the worst-looking point makes the correlation **stronger** — most teams will have predicted that direction. What they will not have predicted is the size: only +0.024. Hold that number for 3.2.

Q3.2: leave-one-out changes, largest first: Onward **+0.0473**, Cars 2 +0.0239, WALL-E +0.0091. Onward is at (88, 61) — middle of the pack horizontally, well below everything else vertically.

Q3.3: the decomposition. r = 0.802 observed; **0.867** if Cars 2 had scored 46; **0.826** with Cars 2 removed.

Where the 0.04 went. Removing Cars 2 does two opposing things. It deletes a point that is out of line, which **raises** r by about 0.065. It also collapses the spread of `rotten_tomatoes` — range 60 down to 31, SD 14.26 down to 9.49 — and r rewards spread, which **lowers** r by about 0.041. Net: +0.024. Onward gets no such cancellation, because at RT = 88 it sits inside the range and removing it changes the spread not at all.

Q3.4: Yes, Cars 2 is genuinely the most unusual point. No, it is not the point that most affects r. Both are true because r is not a measure of unusualness — it asks how tightly the cloud hugs a line, and Cars 2 simultaneously hurts that (off the line) and helps it (widens the spread).

This is where students push hardest, and they are right to. Cars 2 really is the extreme point by every standard diagnostic — its leverage is 0.599 against a mean of 0.095, and its Cook's distance is 3.69 against Onward's 0.147, a factor of 25. If a team argues that Cars 2 "obviously matters more," agree with them. The lesson is not that Cars 2 is unimportant; it is that Δr answers a narrower question than "which point matters," and answering the broader one takes a fitted line.

Hand-off to Class 18. Once regression is on the board, this same scatter gives you residuals and leverage — the tools that do convict Cars 2. Worth promising out loud at the end of Part 3: "the measure that catches this one is next class."

Q3.5: Anscombe's quartet. All four have mean $x = 9$, mean $y = 7.5$, SD $y \approx 1.94$, and $r \approx 0.816$ — close to the 0.802 students just computed, which is the hook. Set I is an ordinary linear scatter; II is a clean parabola; III is a perfect line with one outlier; IV is a vertical stack at $x = 8$ plus a single distant point that manufactures the entire correlation.

Only Set I can honestly be summarized by r. II has a strong relationship that is not linear, so r understates it. III's underlying relationship is nearly perfect but one point drags r down. IV has no relationship at all among ten of its eleven points — and it is the tidiest possible illustration of the Cars 2 problem, since there a single high-leverage point creates the correlation rather than merely surviving it.

Students often conclude "r is useless." Push back: r answers one specific question — how close is this to a straight line — and answers it well. The failure is reporting it without the plot, not the statistic itself.

Two notes on the code

The warmup's `ratings.to_df().dropna()` keeps all 21 films even though Soul is missing a `cinema_score`. That is because `Table.read_table` reads `cinema_score` as a string column, so the missing entry becomes the literal string `'nan'`, which `dropna()` does not recognize. It gives the right answer here, but by luck — reading the same file with `pandas.read_csv` directly drops Soul and leaves 20. Worth knowing if a student's numbers disagree with yours.

The influence loop recomputes standard units inside each subset, which is the correct thing to do — r is defined relative to the mean and SD of the data you actually have. A student who reuses the full-table standard units will get slightly different numbers and is worth a conversation.